

# De-Fi Super-App

A Production-Grade Protocol Ecosystem on Arbitrum Sepolia

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# Modular Ecosystem

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Combining Automated Market Making, Yield Generation,  
and Decentralized Governance into a single reactive interface.

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# Smart Contract Core

## AMMPair Engine

Implemented a custom Constant Product Market Maker (CPMM) with strictly enforced security invariants.

$$x \cdot y = k$$

- **ERC-4626 Vault:** Tokenized shares with precise asset accounting.
- **ReentrancyGuard:** Protection against recursive drain attacks.
- **Slippage Control:** Rigid output validation.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import {ERC20} from "openzeppelin-contracts/contracts/token/ERC20/ERC20.sol";
5 import {IERC20} from "openzeppelin-contracts/contracts/token/ERC20/IERC20.sol";
6 import {SafeERC20} from "openzeppelin-contracts/contracts/token/ERC20/utils/SafeERC20.sol";
7 import {ReentrancyGuard} from "openzeppelin-contracts/contracts/utils/ReentrancyGuard.sol";
8 import {Math} from "openzeppelin-contracts/contracts/utils/math/Math.sol";
9
10 contract AMMPair is ERC20, ReentrancyGuard {
11     using SafeERC20 for IERC20;
12
13     IERC20 public immutable token0;
14     IERC20 public immutable token1;
15
16     uint256 public reserve0;
17     uint256 public reserve1;
18
19     constructor(address _token0, address _token1) ERC20("AMM LP Token", "AMMLP") {
20         token0 = IERC20(_token0);
21         token1 = IERC20(_token1);
22     }
23
24     function addLiquidity(uint256 amount0, uint256 amount1) external nonReentrant returns (uint256 liquidity) {
25         token0.safeTransferFrom(msg.sender, address(this), amount0);
26         token1.safeTransferFrom(msg.sender, address(this), amount1);
27
28         uint256 _totalSupply = totalSupply();
29         if (_totalSupply == 0) {
30             liquidity = Math.sqrt(amount0 * amount1);
31         } else {
32             liquidity = Math.min((amount0 * _totalSupply) / reserve0, (amount1 * _totalSupply) / reserve1);
33         }
34
35         require(liquidity > 0, "Insufficient liquidity minted");
36         _mint(msg.sender, liquidity);
37         _update(token0.balanceOf(address(this)), token1.balanceOf(address(this)));
38     }
39 }
```

# On-Chain Governance



## Protocol Governor

Built from scratch using OpenZeppelin standards.  
Configured with a 4% quorum and 1-week voting period.



## Timelock Controller

A 2-day execution delay prevents malicious admin actions, ensuring community-led security.

# Rigorous Validation

Surpassing project benchmarks with a comprehensive Foundry suite of 83 tests.



*Testing methodology: Randomized Fuzzing + Strict  $\$K\$$ -value Conservation Invariants.*

# Client Architecture

## Reactive Gateway

Built on React 19 + Vite for a high-performance, monochrome production interface.

- **Wagmi v3:** Type-safe Web3 hooks for account synchronization.
- **Viem:** Low-latency RPC transport layer.
- **Reactive State:** Real-time wallet and balance updates.

## ADR #04 Choice

Selected Wagmi over raw Ethers.js to leverage declarative synchronization and native caching, eliminating UI flickering.

# Optimized Data Layer



## The Graph Subgraph

Decoupled historical data retrieval from real-time RPC to ensure sub-second page loads.

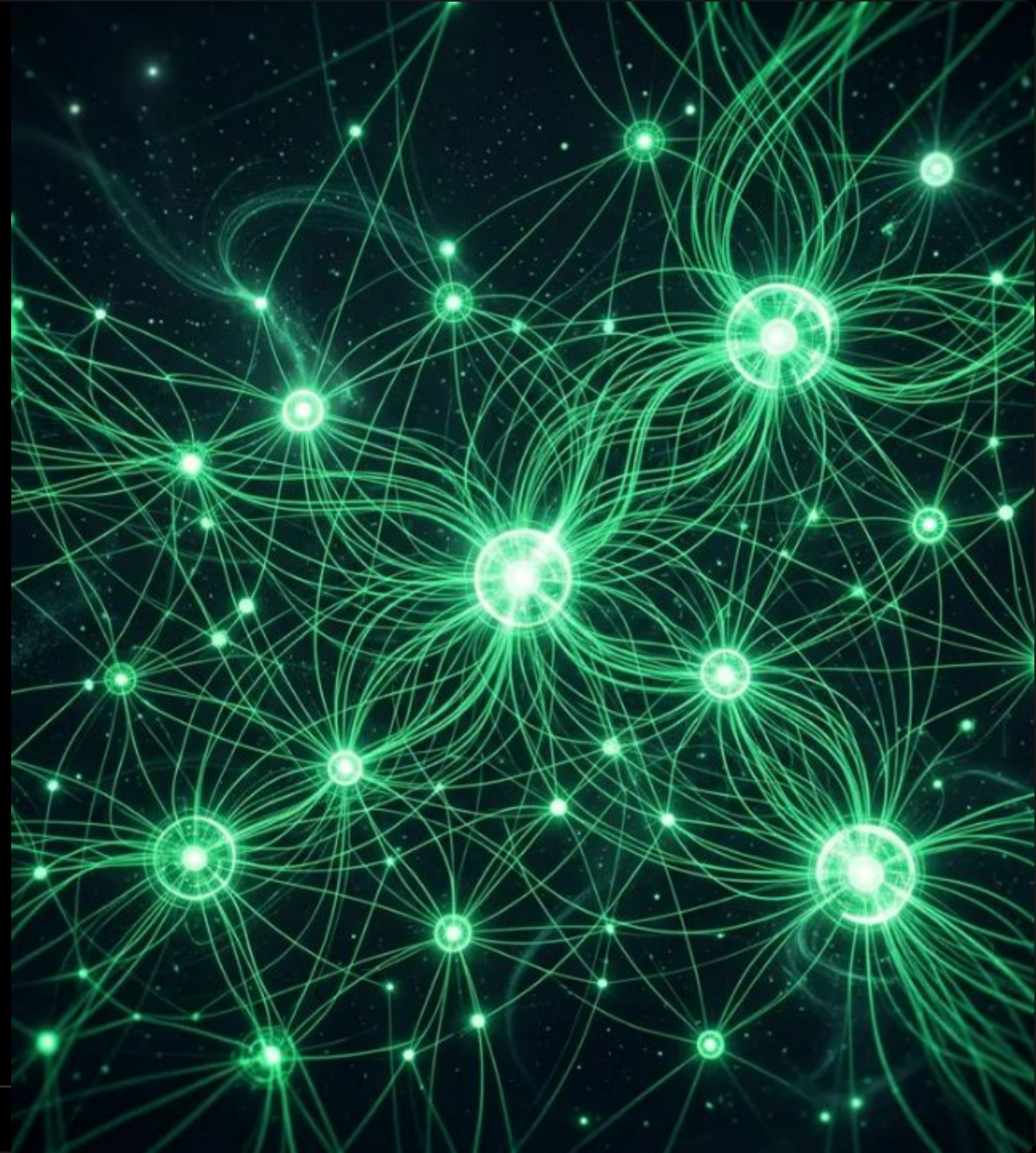
- **Entities:** Normalized relational data for Swaps, Deposits, and Votes.
- **GraphQL:** Complex history filtering (top swaps, proposal life-cycles).
- **Efficiency:** Eliminated expensive L2 node log scanning.



# The Gatekeeper

Built a centralized GitHub Actions CI Pipeline (test.yml) to enforce production quality on every merge.

- **Automated CI:** Triggers on every Push/PR.
- **Branch Protection:** Blocks merging if status is "Red".
- **Slither Audit:** Static security analysis in the cloud.





# Automated Quality Gates



## Slither Security

Compiles AST and runs detectors for Reentrancy and Access Control violations before deployment.



## Linting & Styling

Mandatory **forge fmt** and **Prettier** checks ensure 100% style compliance across teams.

# Infrastructure Rigidity

100%

## Branch Protection Rule

Successfully blocked direct Main pushes.  
Proved via "protected branch hook declined"  
logs. Forced all feature routing through CI  
status checks.

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# Final Defense

Thank you for your attention.

**Deployment:** Arbitrum Sepolia

**Backend:** Solidity ^0.8.20

**Tests:** 83 Pass (Foundry)

**Frontend:** React 19 + Wagmi

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